Article Demo: Using `preamble.tex`, `mathcmd.tex`, `theorem.tex`

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Abstract

This demo exercises most packages and all commands/environments defined in the provided files: `fontspec`, page layout, math tools, figures (`graphicx/subcaption/wrapfig`), TikZ libraries, tables, `hyperref`, and `biblatex`. We cite Lamport (1994) and also (Einstein 1905).

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1 Math macros from `mathcmd.tex`

1.1 Basic sets and constants

这是中文测试。基础数学符号测试。We use $\mathbb{R}, \mathbb{N}, \mathbb{Z}, \mathbb{Q}$ and e, i:

$$\mathrm{e}^{\mathrm{i}\pi} + 1 = 0, \qquad \int_0^1 x^2 \, \mathrm{d}x = \frac{1}{3}.$$

1.2 Vector calculus operators

Let $u(x, y) = x^2 + y^2$ and $\boldsymbol{v}(x, y) = (x, -y)^\top$. Then we can write

$$\nabla u, \quad \nabla \cdot \boldsymbol{v}, \quad \nabla \times \boldsymbol{v}.$$

(Here we mainly show the macros/typesetting;
the actual values depend on the dimension/convention.)

1.3 Function spaces

We reference the function spaces:

$$u \in H^1(\Omega), \quad f \in L^2(\Omega), \quad \boldsymbol{u} \in H(\mathrm{div}, \Omega), \quad \boldsymbol{w} \in H(\mathrm{curl}, \Omega).$$

1.4 Linear algebra operators

Declared operators:

$$\mathrm{rank}(A), \quad \mathrm{tr}(A), \quad \mathrm{diag}(1, 2, 3), \quad \mathrm{trace}(A).$$

Example:

$$A = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} \Rightarrow \mathrm{rank}(A) = 2,\ \mathrm{tr}(A) = 3,\ \mathrm{trace}(A) = 3,\ \det(A) = 2.$$

2 Theorem-like environments from `theorem.tex`

Definition 2.1 (SPD matrix)**.** A matrix $A \in \mathbb{R}^{n \times n}$ is symmetric positive definite (SPD) if $A = A^\top$ and $\boldsymbol{x}^\top A \boldsymbol{x} > 0$ for all $\boldsymbol{x} \neq 0$.

Assumption 2.2. Assume A is SPD.

Lemma 2.3. *If A is SPD, then all eigenvalues of A are positive.*

Theorem 2.4. *If A is SPD, then* $\mathrm{tr}(A) > 0$.

Proof. SPD implies eigenvalues $\lambda_i > 0$. Then $\mathrm{tr}(A) = \sum_i \lambda_i > 0$. For general LaTeX conventions, see Lamport (1994). □

Proposition 2.5. *If A is SPD and $D = \mathrm{diag}(d_1, \ldots, d_n)$ with all $d_i > 0$, then DAD is SPD.*

Corollary 2.6. *If A is SPD, then* $\mathrm{trace}(A^{-1}) > 0$.

Example 2.7. Let $A = \mathrm{diag}(2, 1)$. Then A is SPD and $\mathrm{tr}(A) = 3$.

Remark 2.8. All these environments share the same counter (reset by section),
as configured in `theorem.tex`.

Note 2.9. This document uses `fontspec`, so compile with XeLaTeX or LuaLaTeX.

3 Graphics: `wrapfig`, `subcaption`, `graphicx`

3.1 Wrapfigure demo

This paragraph demonstrates `wrapfig`. Add a few lines of text so the wrapping is clearly visible. Here is some more text to illustrate how the figure wraps around the text. The figure is placed on the right side of the page, and the text flows around it. You can adjust the width of the figure and the placement (left or right) as needed. Continue adding text to see how it interacts with the wrapped figure. This is useful for including small figures or diagrams without interrupting the flow of the text.

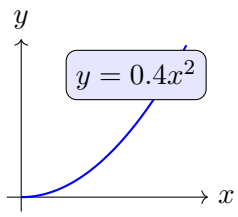


Figure 1: A wrapped TikZ figure.

3.2 Subcaption demo (four figures)



(a) call of duty



(b) f1 the movie



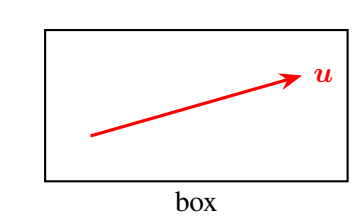
(c) naraka bladepoint



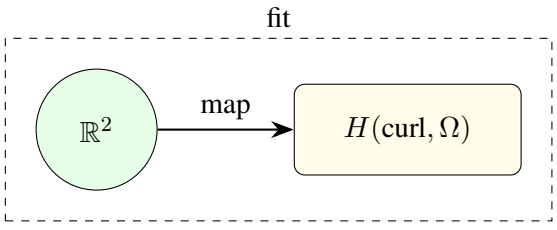
(d) your name shooting

Figure 2: A 2×2 layout of four images

3.3 TikZ libraries demo: calc, patterns, shapes, decorations



(a) Arrow tip from `arrows.meta`.



(b) Positioning + fit.

Figure 3: Two subfigures without external image files.

4 Colors: `xcolor`

We can also show colored text:this is red, this is blue.

5 Tables: `booktabs`, `array`, `multirow`

Item	Example	Note
Trace	$\operatorname{tr}(A)$	operator
	$\operatorname{trace}(A)$	alias
Rank	$\operatorname{rank}(A)$	operator
Space	$H(\operatorname{div}, \Omega),\ H(\operatorname{curl}, \Omega)$	macros

Table 1: A table demonstrating booktabs + array + multirow.

6 Hyperlinks and bibliography

A URL: `https://www.latex-project.org`. A clickable link: CTAN.

We cite (Trefethen and Bau 1997) for numerical linear algebra.

References

Einstein, Albert (1905). “Zur Elektrodynamik bewegter Körper”. In: *Annalen der Physik* 17, pp. 891–921.

Lamport, Leslie (1994). *LaTeX: A Document Preparation System*. Addison-Wesley.

Trefethen, Lloyd N. and David Bau (1997). *Numerical Linear Algebra*. SIAM.